TCN Implementation Example Pytorch

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PCELab
Predictive Computational Engineering

Dataset

Hourly energy demand generation and weather:
From the paper "Tackling Climate Change with Machine Learning", forecasting in energy markets is identified as one of the highest leverage contribution areas of Machine/Deep Learning toward transitioning to a renewable based electrical infrastructure.
This dataset contains 4 years of electrical consumption, generation, pricing, and weather data for Spain. Consumption and generation data was retrieved from ENTSOE a public portal for Transmission Service Operator (TSO) data. Settlement prices were obtained from the Spanish TSO Red Electric España. Weather data was purchased as part of a personal project from the Open Weather API for the 5 largest cities in Spain and made public here.

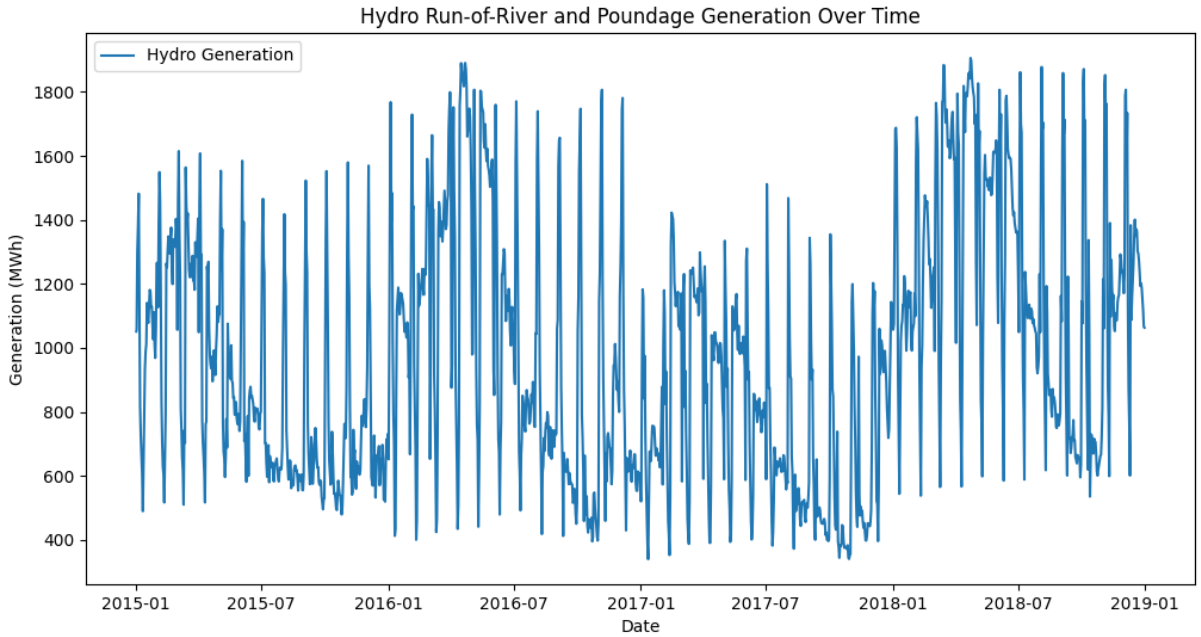


Figure: One week prediction with some more training data

Comparison of TCN Architectures

Table: Encoder Decoder TCN Architecture Summary

Component	Number of Layers	Components per Layer
Chomp1d	1 per TemporalBlock	N/A
TemporalConvNet	3	TemporalBlock
TemporalBlock	2	Conv1d (Weight Norm) Chomp1d ReLU Dropout
TCNEncoder	1	TemporalConvNet (with 1 input channel) Permute Layer
TCNDecoder	3	Conv1d ReLU Dropout Final conv1d

Table: Plain TCN Architecture Summary

Component	Number of Layers	Components per Layer
Chomp1d	1 per TemporalBlock	N/A
TemporalBlock	2	Conv1d (Weight Norm) Chomp1d ReLU Dropout Downsample (Optional) ReLU
TemporalConvNet	3	TemporalBlock
TCN	1	TemporalConvNet Linear Layer

Hyper-parameters

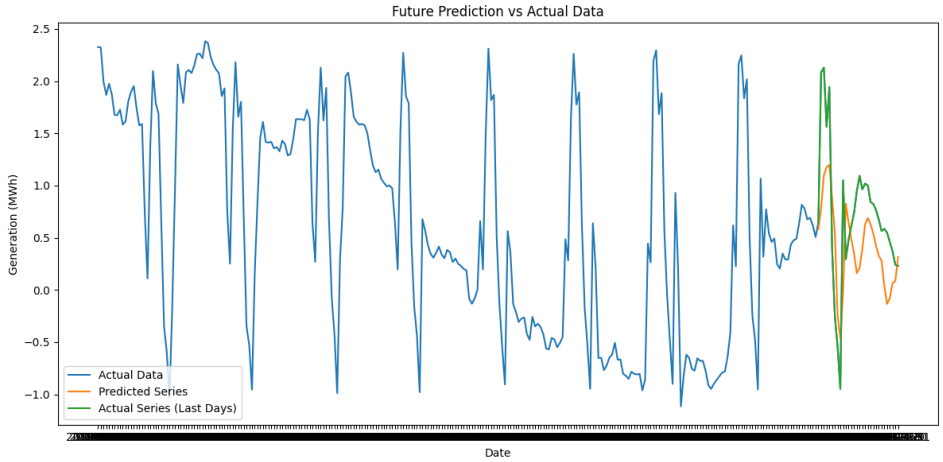
The hyperparameters used for training
Batch size = 16
Dropout ratio= 0.2
Epochs = 10
Kernel (filter) size = 3
Sequence length = 7
Learning rate = 0.001
Optimizer = Adam

Training and Testing

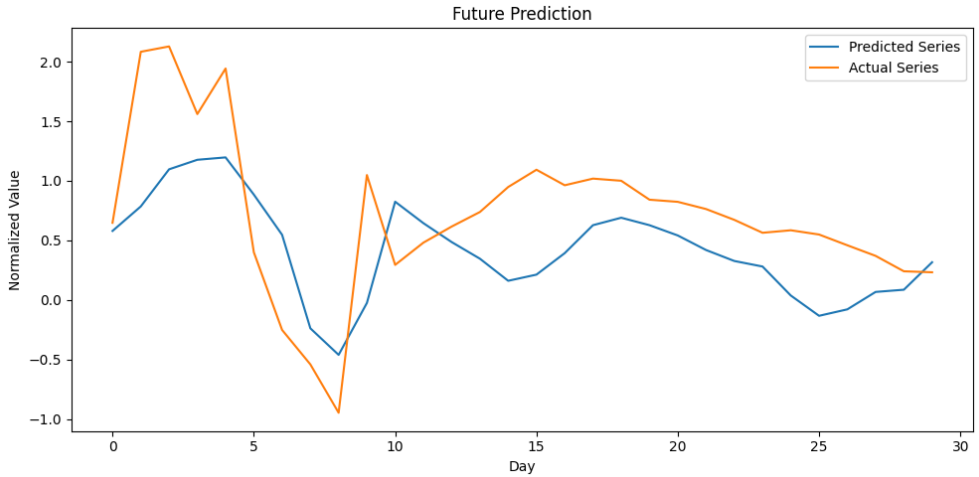
Encoder-Decoder TCN Training and Testing Results
Epoch 1, Loss: 0.6989267638160123
Epoch 2, Loss: 0.1672599609527323
Epoch 3, Loss: 0.11022111835579078
Epoch 4, Loss: 0.0860006758529279
Epoch 5, Loss: 0.07287092336143057
Epoch 6, Loss: 0.06404010077110595
Epoch 7, Loss: 0.05766288755047652
Epoch 8, Loss: 0.05555434017959568
Epoch 9, Loss: 0.052257696270114846
Epoch 10, Loss: 0.049869451568358474
Test set: Average loss: 0.025551
Plain TCN Training and Testing Results
End of Epoch 1 Learning rate: 0.0010 Loss: 0.814014
End of Epoch 2 Learning rate: 0.0010 Loss: 0.483615
End of Epoch 3 Learning rate: 0.0010 Loss: 0.290282
End of Epoch 4 Learning rate: 0.0010 Loss: 0.250012
End of Epoch 5 Learning rate: 0.0010 Loss: 0.234309
End of Epoch 6 Learning rate: 0.0010 Loss: 0.221363
End of Epoch 7 Learning rate: 0.0010 Loss: 0.215366
End of Epoch 8 Learning rate: 0.0010 Loss: 0.209273
End of Epoch 9 Learning rate: 0.0010 Loss: 0.204705
End of Epoch 10 Learning rate: 0.0010 Loss: 0.198154

Test set: Average loss: 0.404127

Model 1: Encoder decoder TCN Results

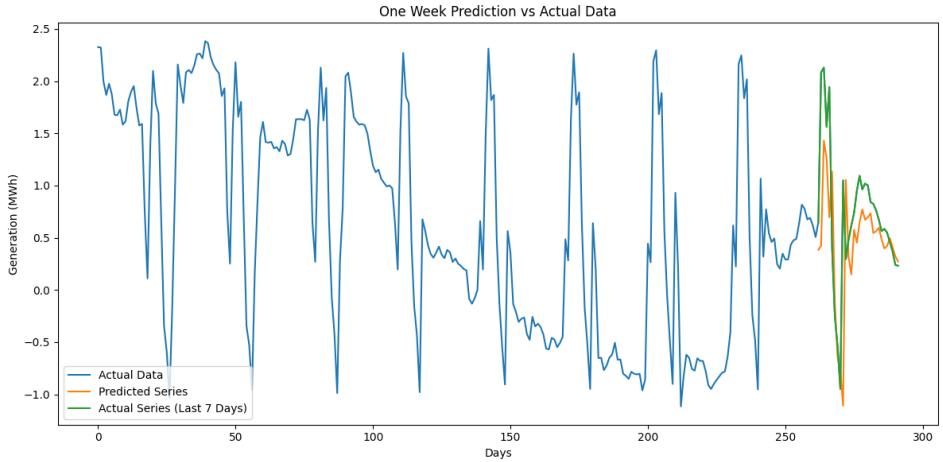


(a) 30 Days prediction with some training data

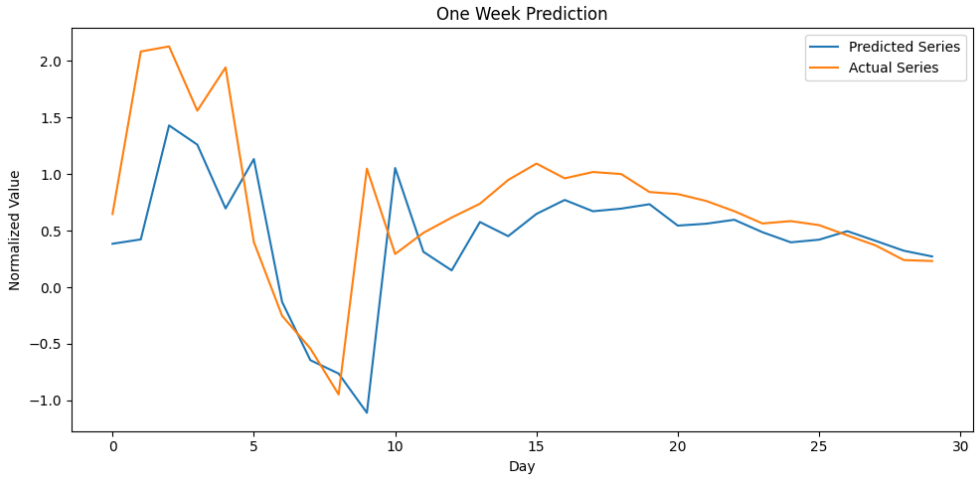


(b) 30 Days Prediction 10 epochs

Model 2: Single TCN Results

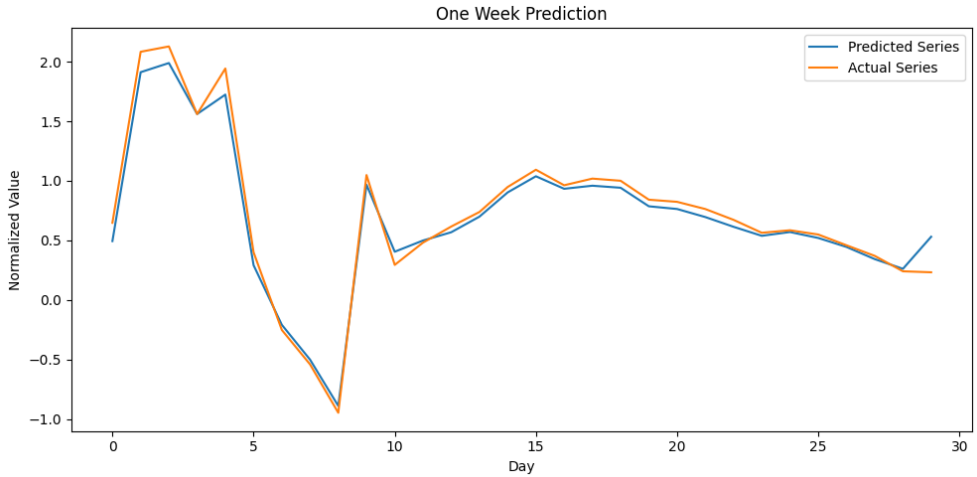


(a) 30 Days prediction with more training data

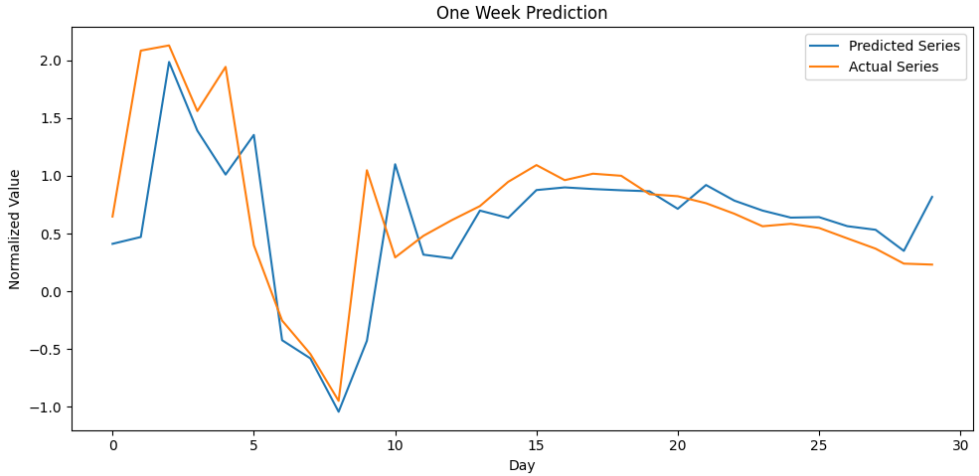


(b) 30 days prediction 10 epochs

Results 100 epochs

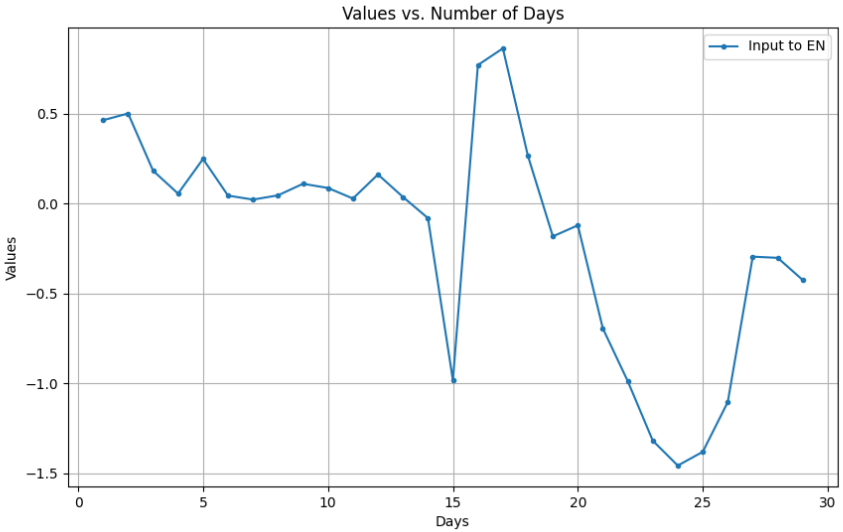


(a) 30 Days prediction with Encoder Decoder

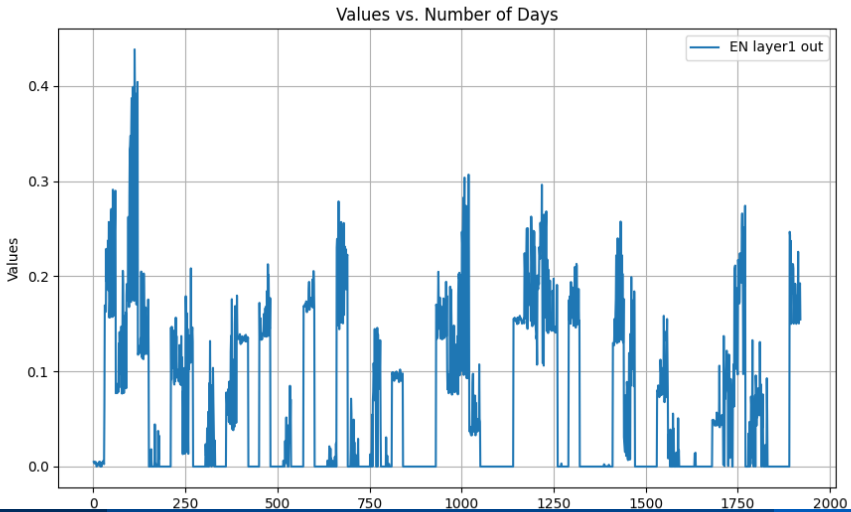


(b) 30 days prediction with plain TCN

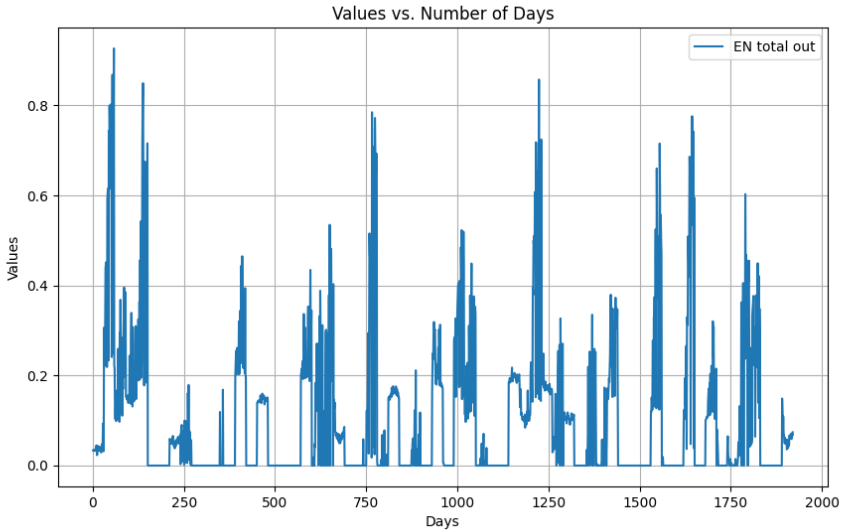
Layer output dissection



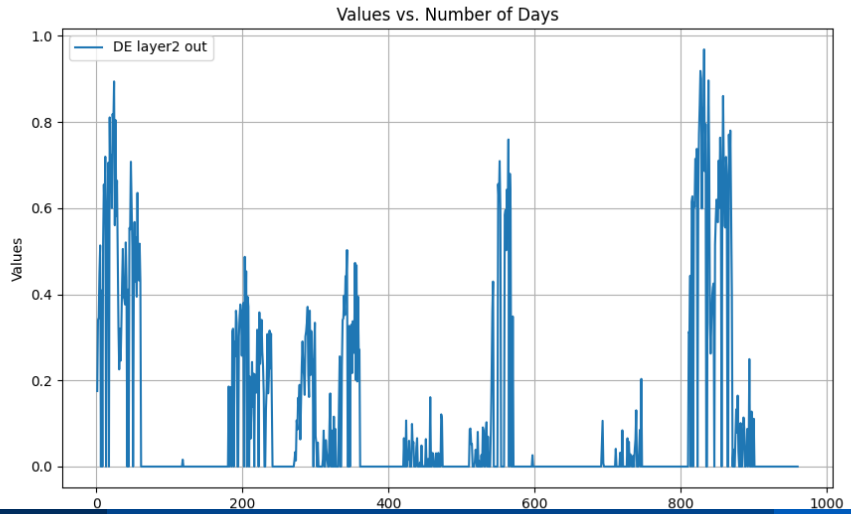
(a) 30 Days prediction with Encoder Decoder



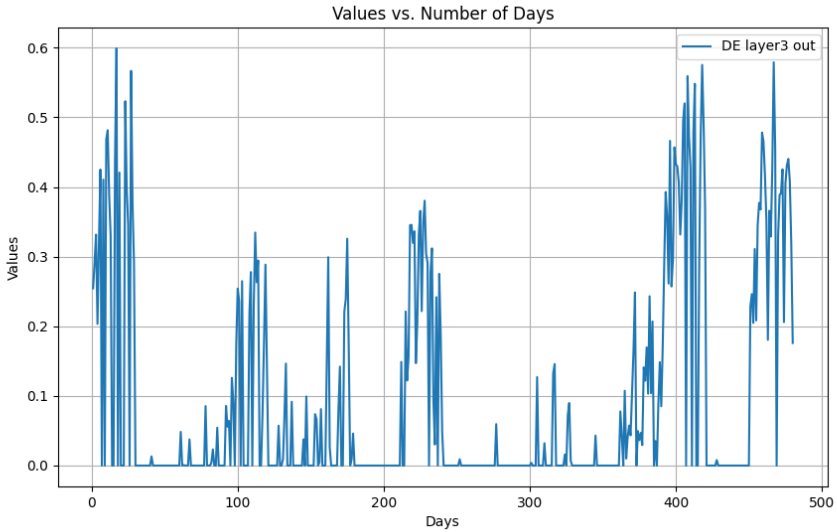
Layer output dissection



(a) 30 Days prediction with Encoder Decoder



Layer output dissection



(a) 30 Days prediction with Encoder Decoder

